

Question ID 4e375d1f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 4e375d1f

How many meters are equivalent to **2,300** centimeters? (**100 centimeters = 1 meter**)

- A. **0.043**
- B. **23**
- C. **2,400**
- D. **230,000**

ID: 4e375d1f Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that **100** centimeters is equal to **1** meter. Therefore, **2,300** centimeters is equivalent to **(2,300 centimeters) $\left(\frac{1 \text{ meter}}{100 \text{ centimeters}}\right)$, or 23** meters.

Choice A is incorrect. **0.043** meters is equivalent to **4.3**, not **2,300**, centimeters.

Choice C is incorrect. **2,400** meters is equivalent to **240,000**, not **2,300**, centimeters.

Choice D is incorrect. **230,000** meters is equivalent to **23,000,000**, not **2,300**, centimeters.

Question Difficulty:
Easy

Question ID 763e6769

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 763e6769

The ratio x to y is equivalent to the ratio 12 to t . When $x = 156$, what is the value of y in terms of t ?

- A. $13t$
- B. $12t$
- C. $144t$
- D. $168t$

ID: 763e6769 Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that the ratio x to y is equivalent to the ratio 12 to t . This can be represented by $\frac{x}{y} = \frac{12}{t}$. Substituting 156 for x in this equation yields $\frac{156}{y} = \frac{12}{t}$. This can be rewritten as $12y = 156t$. Dividing both sides of this equation by 12 yields $y = 13t$. Therefore, when $x = 156$, the value of y in terms of t is $13t$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID 4347a032

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 4347a032

How many teaspoons are equivalent to ~~44~~ tablespoons? (~~3~~ teaspoons = ~~1~~ tablespoon)

- A. ~~47~~
- B. ~~88~~
- C. **132**
- D. ~~176~~

ID: 4347a032 Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that ~~3~~ teaspoons is equivalent to ~~1~~ tablespoon. Therefore, ~~44~~ tablespoons is equivalent to ~~(44 tablespoons)~~ $\left(\frac{3 \text{ teaspoons}}{1 \text{ tablespoon}}\right)$, or **132** teaspoons.

Choice A is incorrect. This is equivalent to approximately **15.66** tablespoons, not ~~44~~ tablespoons.

Choice B is incorrect. This is equivalent to approximately **29.33** tablespoons, not ~~44~~ tablespoons.

Choice D is incorrect. This is equivalent to approximately **58.66** tablespoons, not ~~44~~ tablespoons.

Question Difficulty:
Easy

Question ID d7a3179d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: d7a3179d

How many yards are equivalent to **1,116** inches? (**1 yard = 36 inches**)

ID: d7a3179d Answer

Correct Answer:
31

Rationale

The correct answer is **31**. It's given that **1** yard is equal to **36** inches. Therefore, **1,116** inches is equivalent to **(1,116 inches)** $\left(\frac{1 \text{ yard}}{36 \text{ inches}}\right)$, or **31** yards.

Question Difficulty:
Easy

Question ID 808f7d6c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 808f7d6c

If $t = 4u$, which of the following is equivalent to $2t$?

- A. $8u$
- B. $2u$
- C. u
- D. $\frac{1}{2}u$

ID: 808f7d6c Answer

Correct Answer:
A

Rationale

Choice A is correct. It’s given that $t = 4u$. Multiplying both sides of this equation by 2 yields $2t = 2(4u)$, or $2t = 8u$.

Choice B is incorrect and may result from dividing, instead of multiplying, the right-hand side of the equation by 2. Choices C and D are incorrect and may result from calculation errors.

Question Difficulty:
Easy

Question ID 3318d37b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 3318d37b

A product costs **11.00** dollars per pound. What is the cost, in dollars, for **6** pounds of the product?

ID: 3318d37b Answer

Correct Answer:
66

Rationale

The correct answer is **66**. It’s given that a product costs **11.00** dollars per pound. Therefore, the cost for **6** pounds of the product is $\left(\frac{11.00 \text{ dollars}}{1 \text{ pound}}\right)(6 \text{ pounds})$, which is equivalent to **66.00**, or **66**, dollars.

Question Difficulty:
Easy

Question ID 99550621

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 99550621

Makayla is planning an event in a 5,400-square-foot room. If there should be at least 8 square feet per person, what is the maximum number of people that could attend this event?

- A. 588
- B. 675
- C. 15,274
- D. 43,200

ID: 99550621 Answer

Correct Answer:
B

Rationale

Choice B is correct. It’s given that the event will be in a 5,400-square-foot room and that there should be at least 8 square feet per person. The maximum number of people that could attend the event can be found by dividing the total square feet in the room by the minimum number of square feet needed per person, which gives $\frac{5,400}{8} = 675$.

Choices A and C are incorrect and may result from conceptual or computational errors. Choice D is incorrect and may result from multiplying, rather than dividing, 5,400 by 8.

Question Difficulty:
Easy

Question ID 4837406c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 4837406c

An object travels at a constant speed of **6** centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel **24** centimeters?

ID: 4837406c Answer

Correct Answer:

4

Rationale

The correct answer is **4**. It's given that the object travels at a constant speed of **6** centimeters per second. The speed of the object can be written as $\frac{6 \text{ centimeters}}{1 \text{ second}}$. Let x represent the time, in seconds, it would take for the object to travel **24** centimeters. The value of x can be calculated by solving the equation $\frac{6 \text{ centimeters}}{1 \text{ second}} = \frac{24 \text{ centimeters}}{x \text{ seconds}}$, which can be written as $\frac{6}{1} = \frac{24}{x}$, or $6 = \frac{24}{x}$. Multiplying each side of this equation by x yields $6x = 24$. Dividing each side of this equation by **6** yields $x = 4$. Therefore, it would take the object **4** seconds to travel **24** centimeters.

Question Difficulty:

Easy

Question ID 94660ba8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 94660ba8

A participant in a bicycle race completes the race with an average speed of **24,816** yards per hour. What is this average speed, in miles per hour? (**1 mile = 1,760 yards**)

ID: 94660ba8 Answer

Correct Answer:
14.1

Rationale

The correct answer is **14.1**. It’s given that a participant completes the bicycle race with an average speed of **24,816** yards per hour and **1 mile = 1,760 yards**. It follows that this average speed is equivalent to $\left(\frac{24,816 \text{ yards}}{1 \text{ hour}}\right)\left(\frac{1 \text{ mile}}{1,760 \text{ yards}}\right)$, which yields $\frac{14.1 \text{ miles}}{1 \text{ hour}}$, or **14.1** miles per hour.

Question Difficulty:
Easy

Question ID 551c52b9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 551c52b9

Tilly earns p dollars for every w hours of work. Which expression represents the amount of money, in dollars, Tilly earns for $39w$ hours of work?

- A. $39p$
- B. $\frac{p}{39}$
- C. $p + 39$
- D. $p - 39$

ID: 551c52b9 Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that Tilly earns p dollars for every w hours of work. This can be represented by the proportion $\frac{p}{w}$. The amount of money, x , Tilly earns for $39w$ hours of work can be found by setting up the proportion $\frac{p}{w} = \frac{x}{39w}$. This can be rewritten as $39pw = xw$. Dividing both sides by w results in $x = 39p$.

Choice B is incorrect. This is the amount of money Tilly earns in dollars per hour, not the amount of money Tilly earns for $39w$ hours of work.

Choice C is incorrect. This is the amount of money Tilly earns for w hours of work plus 39 , not the amount of money Tilly earns for $39w$ hours of work.

Choice D is incorrect. This is the amount of money Tilly earns for w hours of work minus 39 , not the amount of money Tilly earns for $39w$ hours of work.

Question Difficulty:
Easy

Question ID c47256ca

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: c47256ca

An object's speed is **64 yards** per second. What is the object's speed, in feet per second? (**1 yard = 3 feet**)

- A. **61**
- B. **67**
- C. **94**
- D. **192**

ID: c47256ca Answer

Correct Answer:
D

Rationale

Choice D is correct. Since **1** yard is equal to **3** feet, **64** yards is equal to **64 yards** $\left(\frac{3 \text{ feet}}{1 \text{ yard}}\right)$, or **192** feet. It follows that **64** yards per second is equivalent to **192** feet per second. Therefore, the object's speed is **192** feet per second.

Choice A is incorrect. A speed of **61** feet per second is equivalent to $\frac{61}{3}$, not **64**, yards per second.

Choice B is incorrect. A speed of **67** feet per second is equivalent to $\frac{67}{3}$, not **64**, yards per second.

Choice C is incorrect. A speed of **94** feet per second is equivalent to $\frac{94}{3}$, not **64**, yards per second.

Question Difficulty:
Easy

Question ID 8bf3f67a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 8bf3f67a

A special camera is used for underwater ocean research. The camera is at a depth of **39** fathoms. What is the camera's depth in feet? (**1 fathom = 6 feet**)

- A. **234**
- B. 117
- C. 45
- D. 7

ID: 8bf3f67a Answer

Correct Answer:
A

Rationale

Choice A is correct. It’s given that a special camera is used for underwater ocean research, and this camera is at a depth of **39** fathoms. It's also given that **1** fathom is equal to **6** feet. Thus, **39** fathoms is equivalent to **(39 fathoms) $(\frac{6 \text{ feet}}{1 \text{ fathom}})$** , or **234** feet. Therefore, the camera's depth, in feet, is **234**.

Choice B is incorrect. This is the camera's depth, in feet, if the camera is at a depth of **19.5** fathoms.

Choice C is incorrect. This is the camera's depth, in feet, if the camera is at a depth of **7.5** fathoms.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID 7270d642

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 7270d642

A certain bird species can fly at an average speed of **16** meters per second when in continuous flight. At this rate, how many meters would this bird species fly in **4** seconds?

- A. **64**
- B. **20**
- C. **16**
- D. **12**

ID: 7270d642 Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that a certain bird species can fly at an average speed of **16** meters per second when in continuous flight. At this rate, in **4** seconds this bird species would fly $(\frac{16 \text{ meters}}{\text{second}})(4 \text{ seconds})$, or **64** meters.

Choice B is incorrect. This is the value of **16 + 4**, not **16(4)**.

Choice C is incorrect. This is the distance the bird would fly in **1** second, not **4** seconds.

Choice D is incorrect. This is the value of **16 - 4**, not **16(4)**.

Question Difficulty:
Easy

Question ID 3ac09984

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 3ac09984

Marta has 7,500 pesos she will convert to US dollars using a currency exchange service. At this time, the currency exchange rate is 1 peso = 0.075 US dollars. The exchange service will charge Marta a 2% fee on the converted US dollar amount. How many US dollars will Marta receive from the currency exchange after the 2% fee is applied?

- A. \$551.25
- B. \$562.50
- C. \$5,625.00
- D. \$98,000.00

ID: 3ac09984 Answer

Correct Answer:

A

Rationale

Choice A is correct. At the exchange rate of 1 peso = 0.075 US dollars, 7,500 pesos would be converted to $7,500 \times 0.075 = \$562.50$. However, since Maria pays a 2% fee on the converted US dollar amount, she receives only $(100 - 2)\%$, or 98%, of the converted US dollars, and $562.50 \times 0.98 = \$551.25$.

Choice B is incorrect. This is the number of US dollars Maria would receive if the exchange service did not charge a 2% fee. Choice C is incorrect and may result from a decimal point error made when calculating the conversion to US dollars and from not assessing the 2% fee. Choice D is incorrect and may result from reversing the units of the exchange rate.

Question Difficulty:

Easy

Question ID 2cdefcb1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 2cdefcb1

What length, in centimeters, is equivalent to a length of **51** meters? (**1 meter = 100 centimeters**)

- A. 0.051
- B. 0.51
- C. 5,100
- D. 51,000

ID: 2cdefcb1 Answer

Correct Answer:
C

Rationale

Choice C is correct. Since **1** meter is equal to **100** centimeters, **51** meters is equal to **51 meters**($\frac{100 \text{ centimeters}}{1 \text{ meter}}$), or **5,100** centimeters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from dividing, rather than multiplying, **51** by **100**.

Choice D is incorrect. This is the length, in millimeters rather than centimeters, that is equivalent to a length of **51** meters.

Question Difficulty:
Easy

Question ID 85b33aa8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 85b33aa8

A fish swam a distance of **5,104** yards. How far did the fish swim, in miles? (**1 mile = 1,760 yards**)

- A. **0.3**
- B. **2.9**
- C. **3,344**
- D. **6,864**

ID: 85b33aa8 Answer

Correct Answer:

B

Rationale

Choice B is correct. It’s given that the fish swam **5,104** yards and that **1** mile is equal to **1,760** yards. Therefore, the fish swam **5,104 yards** $\left(\frac{1 \text{ mile}}{1,760 \text{ yards}}\right)$, which is equivalent to $\frac{5,104}{1,760}$ **miles**, or **2.9** miles.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID 06a152cd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 06a152cd

To make a bakery’s signature chocolate muffins, a baker needs 2.5 ounces of chocolate for each muffin. How many pounds of chocolate are needed to make 48 signature chocolate muffins? (1 pound = 16 ounces)

- A. 7.5
- B. 10
- C. 50.5
- D. 120

ID: 06a152cd Answer

Correct Answer:

A

Rationale

Choice A is correct. If 2.5 ounces of chocolate are needed for each muffin, then the number of ounces of chocolate needed to make 48 muffins is $48 \times 2.5 = 120$ ounces. Since 1 pound = 16 ounces, the number of pounds that is equivalent to 120 ounces is $\frac{120}{16} = 7.5$ pounds. Therefore, 7.5 pounds of chocolate are needed to make the 48 muffins.

Choice B is incorrect. If 10 pounds of chocolate were needed to make 48 muffins, then the total number of ounces of chocolate needed would be $10 \times 16 = 160$ ounces. The number of ounces of chocolate per muffin would then be $\frac{160}{48} = 3.33$ ounces per muffin, not 2.5 ounces per muffin. Choices C and D are also incorrect. Following the same procedures as used to test choice B gives 16.8 ounces per muffin for choice C and 40 ounces per muffin for choice D, not 2.5 ounces per muffin. Therefore, 50.5 and 120 pounds cannot be the number of pounds needed to make 48 signature chocolate muffins.

Question Difficulty:

Easy

Question ID d0d9ede4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: d0d9ede4

How many feet are equivalent to **34** yards? (**1 yard = 3 feet**)

ID: d0d9ede4 Answer

Correct Answer:
102

Rationale

The correct answer is **102**. It's given that **1** yard is equivalent to **3** feet. Therefore, **34** yards is equivalent to **(34 yards)** $\left(\frac{3\text{ feet}}{1\text{ yard}}\right)$, or **102** feet.

Question Difficulty:
Easy

Question ID 24ad9dcb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 24ad9dcb

The weight of an object on Venus is approximately $\frac{9}{10}$ of its weight on Earth. The weight of an object on Jupiter is approximately $\frac{23}{10}$ of its weight on Earth. If an object weighs 100 pounds on Earth, approximately how many more pounds does it weigh on Jupiter than it weighs on Venus?

- A. 90
- B. 111
- C. 140
- D. 230

ID: 24ad9dcb Answer

Correct Answer:

C

Rationale

Choice C is correct. The weight of an object on Venus is approximately $\frac{9}{10}$ of its weight on Earth. If an object weighs 100 pounds on Earth, then the object’s weight on Venus is approximately $\frac{9}{10}(100) = 90$ pounds. The same object’s weight on Jupiter is approximately $\frac{23}{10}$ of its weight on Earth; therefore, the object weighs approximately $\frac{23}{10}(100) = 230$ pounds on Jupiter. The difference between the object’s weight on Jupiter and the object’s weight on Venus is approximately $230 - 90 = 140$ pounds. Therefore, an object that weighs 100 pounds on Earth weighs 140 more pounds on Jupiter than it weighs on Venus.

Choice A is incorrect because it is the weight, in pounds, of the object on Venus. Choice B is incorrect because it is the weight, in pounds, of an object on Earth if it weighs 100 pounds on Venus. Choice D is incorrect because it is the weight, in pounds, of the object on Jupiter.

Question Difficulty:

Easy

Question ID 7cd1c6db

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 7cd1c6db

An object travels at a constant speed of **12** centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel **108** centimeters?

- A. **9**
- B. **96**
- C. **120**
- D. **972**

ID: 7cd1c6db Answer

Correct Answer:
A

Rationale

Choice A is correct. If the object travels **108** centimeters at a speed of **12** centimeters per second, the time of travel can be determined by dividing the total distance by the speed. This results in $\frac{108 \text{ centimeters}}{12 \text{ centimeters/second}}$, which is **9** seconds.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID ba62b0b0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: ba62b0b0

A kangaroo has a mass of **28** kilograms. What is the kangaroo's mass, in grams? (**1 kilogram = 1,000 grams**)

- A. **28,000**
- B. **1,028**
- C. **972**
- D. **784**

ID: ba62b0b0 Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that a kangaroo has a mass of **28** kilograms and that **1** kilogram is equal to **1,000** grams. Therefore, the kangaroo's mass, in grams, is **28 kilograms** $\left(\frac{1,000 \text{ grams}}{1 \text{ kilogram}}\right)$, which is equivalent to **28,000** grams.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID fe1ec415

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: fe1ec415

A cherry pitting machine pits **12** pounds of cherries in **3** minutes. At this rate, how many minutes does it take the machine to pit **96** pounds of cherries?

- A. **8**
- B. **15**
- C. **24**
- D. **36**

ID: fe1ec415 Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that the cherry pitting machine pits **12** pounds of cherries in **3** minutes. This rate can be written as $\frac{12 \text{ pounds of cherries}}{3 \text{ minutes}}$. If the number of minutes it takes the machine to pit **96** pounds of cherries is represented by x , the value of x can be calculated by solving the equation $\frac{12 \text{ pounds of cherries}}{3 \text{ minutes}} = \frac{96 \text{ pounds of cherries}}{x \text{ minutes}}$, which can be rewritten as $\frac{12}{3} = \frac{96}{x}$, or $4 = \frac{96}{x}$. Multiplying each side of this equation by x yields $4x = 96$. Dividing each side of this equation by **4** yields $x = 24$. Therefore, it takes the machine **24** minutes to pit **96** pounds of cherries.

Choice A is incorrect. This is the number of minutes it takes the machine to pit **32**, not **96**, pounds of cherries.

Choice B is incorrect. This is the number of minutes it takes the machine to pit **60**, not **96**, pounds of cherries.

Choice D is incorrect. This is the number of minutes it takes the machine to pit **144**, not **96**, pounds of cherries.

Question Difficulty:
Easy

Question ID 0700a2d5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 0700a2d5

How many yards are equivalent to **77** rods? (**5.5 yards = 1 rod**)

ID: 0700a2d5 Answer

Correct Answer:

423.5, 847/2

Rationale

The correct answer is **423.5**. It's given that **5.5 yards = 1 rod**. Therefore, **77** rods is equivalent to **(77 rods)** $\left(\frac{5.5 \text{ yards}}{1 \text{ rod}}\right)$, or **423.5** yards. Note that 423.5 and 847/2 are examples of ways to enter a correct answer.

Question Difficulty:

Easy

Question ID c81499e1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: c81499e1

A giant armadillo has a mass of **39** kilograms. What is the giant armadillo's mass in **grams**? (**1 kilogram = 1,000 grams**)

ID: c81499e1 Answer

Correct Answer:
39000

Rationale

The correct answer is **39,000**. It's given that the giant armadillo has a mass of **39** kilograms. Since **1** kilogram is equal to **1,000** grams, **39** kilograms is equal to **39 kilograms** $\left(\frac{1,000 \text{ grams}}{1 \text{ kilogram}}\right)$, or **39,000** grams. Therefore, the giant armadillo's mass, in grams, is **39,000**.

Question Difficulty:
Easy

Question ID e9841407

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: e9841407

Shaquan has 7 red cards and 28 blue cards. What is the ratio of red cards to blue cards that Shaquan has?

- A. 1 to 4
- B. 4 to 1
- C. 1 to 7
- D. 7 to 1

ID: e9841407 Answer

Correct Answer:
A

Rationale

Choice A is correct. It’s given that Shaquan has 7 red cards and 28 blue cards. Therefore, the ratio of red cards to blue cards that Shaquan has is 7 to 28. This ratio can be reduced by dividing both parts of the ratio by 7, which yields the ratio 1 to 4.

Choice B is incorrect. This is the ratio of blue cards to red cards that Shaquan has. Choice C is incorrect and may result from a calculation error when reducing the ratio. Choice D is incorrect. This may result from finding the ratio of blue cards to red cards, or 28 to 7, and then making a calculation error when reducing the ratio.

Question Difficulty:
Easy

Question ID aeeaec96

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: aeeaec96

How many yards are equivalent to **612 inches**? (**1 yard = 36 inches**)

- A. **0.059**
- B. **17**
- C. **576**
- D. **22,032**

ID: aeeaec96 Answer

Correct Answer:

B

Rationale

Choice B is correct. It's given that **1 yard = 36 inches**. Therefore, **612 inches** is equivalent to **612 inches** $\left(\frac{1 \text{ yard}}{36 \text{ inches}}\right)$, which can be rewritten as $\frac{612 \text{ yards}}{36}$, or **17 yards**.

Choice A is incorrect. This is the number of yards that are equivalent to **2.124 inches**.

Choice C is incorrect. This is the number of yards that are equivalent to **20,736 inches**.

Choice D is incorrect. This is the number of yards that are equivalent to **793,152 inches**.

Question Difficulty:

Easy

Question ID 2d16d62c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 2d16d62c

A special camera is used for underwater ocean research. When the camera is at a depth of **58** fathoms, what is the camera's depth in feet? (**1 fathom = 6 feet**)

ID: 2d16d62c Answer

Correct Answer:

348

Rationale

The correct answer is **348**. It's given that **1** fathom is equivalent to **6** feet. Therefore, **58** fathoms is equivalent to **(58 fathoms)** $\left(\frac{6 \text{ feet}}{1 \text{ fathom}}\right)$, or **348** feet. Thus, when the camera is at a depth of **58** fathoms, the camera's depth, in feet, is **348**.

Question Difficulty:

Easy

Question ID 6310adbc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 6310adbc

The ratio of t to u is 1 to 2, and $t = 10$.

What is the value of u ?

- A. 2
- B. 5
- C. 10
- D. 20

ID: 6310adbc Answer

Correct Answer:

D

Rationale

Choice D is correct. It’s given that the ratio of t to u is 1 to 2. Since $t = 10$, it follows that the ratio of 10 to u is also 1 to 2. The relationship between these ratios can be represented by the proportion $\frac{10}{u} = \frac{1}{2}$. Multiplying both sides of this equation by 2 and then by u yields $20 = u$.

Choice A is incorrect. This is the value of u when $t = 1$. Choice B is incorrect. This would be the value of u if the ratio of t to u were 2 to 1. Choice C is incorrect. This is the value of t, not u.

Question Difficulty:

Easy

Question ID da9ffcf6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: da9ffcf6

The ratio of the length of line segment XY to the length of line segment ZV is **6 to 1**. If the length of line segment XY is **102** inches, what is the length, in inches, of line segment ZV ?

- A. **17**
- B. **96**
- C. **102**
- D. **612**

ID: da9ffcf6 Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that the ratio of the length of line segment XY to the length of line segment ZV is **6 to 1**, which means $\frac{XY}{ZV} = \frac{6}{1}$. It's given that the length of line segment XY is **102** inches. If the length, in inches, of line segment ZV is represented by ℓ , the value of ℓ can be calculated by solving the equation $\frac{102}{\ell} = \frac{6}{1}$, or $\frac{102}{\ell} = 6$. Multiplying each side of this equation by ℓ yields **102 = 6 ℓ** . Dividing each side of this equation by **6** yields **17 = ℓ** . Therefore, the length of line segment ZV is **17** inches.

Choice B is incorrect. This is the length, in inches, of line segment ZV if the length of line segment XY is **576**, not **102**, inches.

Choice C is incorrect. This is the length, in inches, of line segment XY , not line segment ZV .

Choice D is incorrect. This is the length, in inches, of line segment ZV if the ratio of the length of line segment XY to the length of line segment ZV is **1 to 6**, not **6 to 1**.

Question Difficulty:
Easy

Question ID 3f236a64

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 3f236a64

x	y
1	4
3	12
5	20
40	k

In the table above, the ratio of y to x for each ordered pair is constant. What is the value of k ?

- A. 28
- B. 36
- C. 80
- D. 160

ID: 3f236a64 Answer

Correct Answer:
D

Rationale

Choice D is correct. Since the ratio of y to x is constant for each ordered pair in the table, the first row can be used to determine that the ratio of y to x is 4 to 1. The proportion $\frac{4}{1} = \frac{k}{40}$ can be used to solve for k. Multiplying each side of the equation by 40 yields $160 = k$.

Choice A is incorrect. This is the value of y when the value of x is 7, not 40. Choice B is incorrect and may result from subtracting 4 from 40 instead of multiplying 40 by 4. Choice C is incorrect and may result from incorrectly setting up the proportion.

Question Difficulty:
Easy

Question ID e3d49511

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: e3d49511

The number of raccoons in a **131**-square-mile area is estimated to be **2,358**. What is the estimated population density, in raccoons per square mile, of this area?

- A. **18**
- B. **131**
- C. **149**
- D. **2,376**

ID: e3d49511 Answer

Correct Answer:
A

Rationale

Choice A is correct. It’s given that there are **2,358** raccoons in a **131**-square-mile area. The estimated population density, in raccoons per square mile, is the estimated number of raccoons divided by the number of square miles. Therefore, the estimated population density of this area is $\frac{2,358 \text{ raccoons}}{131 \text{ square miles}}$, or **18** raccoons per square mile.

Choice B is incorrect. This is the number of square miles in the area, not the estimated number of raccoons per square mile in this area.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID be35c117

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: be35c117

A wind turbine completes **900** revolutions in **50** minutes. At this rate, how many revolutions per minute does this turbine complete?

- A. **18**
- B. **850**
- C. **950**
- D. **1,400**

ID: be35c117 Answer

Correct Answer:
A

Rationale

Choice A is correct. Dividing the number of revolutions by the number of minutes gives the number of revolutions the turbine completes per minute. It’s given that the wind turbine completes **900** revolutions in **50** minutes. Therefore, at this rate, this turbine completes $\frac{900}{50}$, or **18**, revolutions per minute.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID 15617f62

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 15617f62

The population density of Worthington is **290** people per square mile. Worthington has a population of **92,800** people. What is the area, in square miles, of Worthington?

- A. **102,400**
- B. **93,090**
- C. **320**
- D. **32**

ID: 15617f62 Answer

Correct Answer:
C

Rationale

Choice C is correct. It’s given that the population density of Worthington is **290** people per square mile and Worthington has a population of **92,800** people. Therefore, the area of Worthington is **92,800 people** $\left(\frac{1 \text{ square mile}}{290 \text{ people}}\right)$, which is equivalent to $\frac{92,800 \text{ square miles}}{290}$, or **320** square miles.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID 312ba47c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 312ba47c

In a box of pens, the ratio of black pens to red pens is **8** to **1**. There are **40** black pens in the box. How many red pens are in the box?

- A. **5**
- B. **8**
- C. **40**
- D. **320**

ID: 312ba47c Answer

Correct Answer:
A

Rationale

Choice A is correct. It’s given that the ratio of black pens to red pens is **8** to **1**. Therefore, there are $\frac{1}{8}$ as many red pens as black pens in the box. It’s also given that there are **40** black pens in the box. Therefore, the number of red pens is $\frac{1}{8}$ of the **40** black pens. Thus, the number of red pens is $40\left(\frac{1}{8}\right)$, or **5**.

Choice B is incorrect. This is the number of black pens in the box for every red pen.

Choice C is incorrect. This is the number of black pens in the box.

Choice D is incorrect. This is the number of red pens in the box if the ratio of black pens to red pens is **1** to **8**.

Question Difficulty:
Easy

Question ID 000259aa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 000259aa

A group of monarch butterflies migrated from Chicago, Illinois, to Michoacán, Mexico, flying a total of 2,100 miles. It took a single butterfly in the group 120 days to travel this route one way. On average, how many miles did the butterfly travel per day?

- A. 0.057
- B. 0.729
- C. 17.5
- D. 24

ID: 000259aa Answer

Rationale

Choice C is correct. If the butterfly traveled 2,100 miles in 120 days, then it traveled, on average, $\frac{2,100 \text{ miles}}{120 \text{ days}} = 17.5$ miles per day.

Choice A is incorrect. This is approximately the average amount of time, in days, it took the butterfly to fly one mile: $\frac{120 \text{ days}}{2,100 \text{ miles}} = 0.057$ days per mile. Choice B is incorrect and may result from an arithmetic error. Choice D is incorrect. This is the number of hours in a day rather than the number of miles flown per day.

Question Difficulty:

Easy

Question ID fe4c1c9e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: fe4c1c9e

A mechanical device in a workshop produces items at a constant rate of **60** items per hour. At this rate, how many items will the mechanical device produce in **3** hours?

ID: fe4c1c9e Answer

Correct Answer:
180

Rationale

The correct answer is **180**. It's given that a mechanical device produces items at a constant rate of **60** items per hour. This rate can be written as $\frac{60 \text{ items}}{1 \text{ hour}}$. Let x represent the number of items the mechanical device will produce in **3** hours at the given rate. It follows that $\frac{60 \text{ items}}{1 \text{ hour}} = \frac{x \text{ items}}{3 \text{ hours}}$, which can be written as $\frac{60}{1} = \frac{x}{3}$, or $60 = \frac{x}{3}$. Multiplying each side of this equation by **3** yields $180 = x$. Therefore, at the given rate, the mechanical device will produce **180** items in **3** hours.

Alternate approach: It's given that a mechanical device produces items at a constant rate of **60** items per hour. At this rate, the mechanical device will produce $(\frac{60 \text{ items}}{1 \text{ hour}})(3 \text{ hours})$, or **180** items in **3** hours.

Question Difficulty:
Easy

Question ID 86636d8f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 86636d8f

A customer spent **\$27** to purchase oranges at **\$3** per pound. How many pounds of oranges did the customer purchase?

ID: 86636d8f Answer

Correct Answer:

9

Rationale

The correct answer is **9**. It’s given that the customer spent **\$27** to purchase oranges at **\$3** per pound. Therefore, the number of pounds of oranges the customer purchased is **$\$27\left(\frac{1\text{ pound}}{\$3}\right)$** , or **9** pounds.

Question Difficulty:

Easy

Question ID 3f5398a6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 3f5398a6

For a person m miles from a flash of lightning, the length of the time interval from the moment the person sees the lightning to the moment the person hears the thunder is k seconds. The ratio of m to k can be estimated to be 1 to 5. According to this estimate, the person is how many miles from a flash of lightning if the time interval is 25 seconds?

- A. 10
- B. 9
- C. 6
- D. 5

ID: 3f5398a6 Answer

Rationale

Choice D is correct. It’s given that the ratio of m to k is estimated to be 1 to 5. Therefore, when $k = 25$, the relationship between these ratios can be expressed by the proportion $\frac{m}{25} = \frac{1}{5}$. Multiplying both sides of this equation by 25 yields $m = 5$.

Choices A, B, and C are incorrect and may result from calculation errors.

Question Difficulty:
Easy

Question ID 3c8fdc40

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 3c8fdc40

A printer produces posters at a constant rate of **42** posters per minute. At what rate, in posters per hour, does the printer produce the posters?

ID: 3c8fdc40 Answer

Correct Answer:
2520

Rationale

The correct answer is **2,520**. There are **60** minutes in one hour. At a rate of **42** posters per minute, the number of posters produced in one hour can be determined by $\left(\frac{42 \text{ posters}}{1 \text{ minute}}\right)\left(\frac{60 \text{ minutes}}{1 \text{ hour}}\right)$, which is **2,520** posters per hour.

Question Difficulty:
Easy